

Invitation to 't Hooft anomalies

Zhengyan Darius Shi

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1 Why anomalies?

The ultimate goal in condensed matter physics is to understand how exotic intrinsically many-body phenomena can emerge from the collective dynamics of a large number of elementary degrees of freedom. Broadly speaking, there are three philosophically distinct strategies towards achieving this goal:

1. Model-specific: distill the essential physics into a concrete model Hamiltonian and (try to) solve the Hamiltonian using whatever methods one can dream up. Famous success stories include Laughlin's wavefunction for the ideal pseudopotential Hamiltonian of fractional quantum Hall states, Bethe's solution of 1+1D integrable spin chains etc.
2. Model-agnostic: on the opposite extreme, we can forget about models completely and enumerate all possible emergent phases of matter. This approach is the driving force behind the classification of topological phases in various dimensions, including spin liquids, fractional quantum Hall states, topological insulators etc. These classification results provide lookup tables: if we find something exotic in experiments, we can search in the lookup table and find out the corresponding phase of matter. But we cannot predict anything about a specific model of interest.
3. Model-aware: partial information about the microscopic model is used to constrain the possible emergent phases in the low energy limit. Classic examples: identify the symmetry of the Hamiltonian and use the symmetry to derive selection rules, ground state degeneracy, symmetry breaking patterns and structure of topological defects etc.

The subject of 't Hooft anomalies belongs to the third category. The basic observation that underpins the subject is that *not all symmetries associated with a group G are created equal*. More precisely, the complete set of symmetry constraints are not just determined by the symmetry group G but also by how the group G acts on local degrees of freedom. 't Hooft anomalies arise when this **symmetry action is “anomalous”** under a suitable definition. It turns out that identifying anomalous symmetries can provide powerful non-perturbative constraints on the low energy dynamics of many condensed matter systems. The goal of these lectures is to explain the

simplest settings in which 't Hooft anomalies arise and derive some of their striking consequences. Towards the end, we will briefly mention some modern developments that challenge our current understanding of anomalies and provide some references for your amusement.

2 Historical baggage: defining an 't Hooft anomaly

Given any symmetry G acting on a quantum mechanical system, a powerful way to probe the anomalous action of G is to couple the system to background G -gauge fields. The partition function of the system in the presence of a background G gauge field A_G is denoted

$$Z[A_G] = \int D\Phi \exp\{iS[\Phi, A_G]\} = \exp\{iS_{\text{eff}}[A_G]\}, \quad (1)$$

where $D\Phi$ denotes a path integral over Φ , which could depend on $d+1$ spacetime coordinates. The case $d = 0$ corresponds to ordinary quantum mechanics, while $d > 0$ corresponds to a quantum field theory in d spatial dimensions.

This background field coupling should be very familiar to every condensed matter physicist. Basically every system we study has a global $U(1)$ symmetry associated with the conservation of electric charge. In this context, a background $U(1)$ gauge field A is nothing but the electromagnetic vector potential. In the presence of a background A , we simply replace ordinary derivatives in the action with covariant derivatives $\partial_\mu \Phi \rightarrow D_\mu \Phi = (\partial_\mu - iA_\mu)\Phi$.¹ Experimentally relevant linear response functions such as conductivity, compressibility, and polarizability are then obtained by expanding $S_{\text{eff}}[A]$ to quadratic order in A .

Naively, one might think that $Z[A_G]$ must be gauge-invariant because it is built out of covariant derivatives. If this is the case, then we can produce a dynamical G -gauge theory by integrating over all gauge configurations

$$Z_{\text{gauged}} = \int DA_G \exp\{iS_{\text{eff}}[A_G]\}. \quad (2)$$

We say that this new theory is obtained from the original theory by **gauging the global symmetry** G . Every physical observable in the new theory must be gauge-invariant.

Remarkably, it turns out that not all symmetries are gaugeable. The ones which are not gaugeable are referred to as **anomalous symmetries**. This leads us to the formal definition of 't Hooft anomalies [1].

't Hooft anomaly: Given a quantum system with a global symmetry G , we say that the system has a G -anomaly if there is an obstruction to gauging the G symmetry. In other words, the partition function $Z[A_G]$ in the presence of background G gauge fields *cannot be made gauge invariant*.

¹This procedure works when the symmetry acts linearly on Φ . In nonlinear sigma models, symmetries generally do not act linearly on the fields and we need a suitably generalized notion of “covariant derivative”.

Two important remarks can be made at this stage:

1. The 't Hooft anomaly is detected by the failure of $Z[A_G]$ to be gauge-invariant, rather than the failure of $S_{\text{eff}}[A_G]$ to be gauge invariant. We will encounter situations in which $S_{\text{eff}}[A_G]$ shifts by $2\pi\mathbb{Z}$ under gauge transformations. In these cases, $S_{\text{eff}}[A_G]$ is manifestly not gauge-invariant but $Z[A_G]$ is. Therefore, 't Hooft anomalies are intrinsically quantum mechanical and not present in a classical theory with the same action.
2. Sometimes, the failure of $Z[A_G]$ to be gauge-invariant can be fixed by local counterterms that depend only on A_G . So the 't Hooft anomaly is only nontrivial when it cannot be removed by any local counterterms. We will see explicit examples of such local counterterms later.

The classification of 't Hooft anomalies in d dimensions is intimately related to the classification of symmetry-protected topological (SPT) phases in $d + 1$ dimensions through the mechanism of **anomaly inflow**. Roughly speaking, the statement is that a failure of gauge invariance for $Z[A_G]$ on a d dimensional closed manifold M_d can always be cured by including a suitable SPT on a $d + 1$ dimensional manifold Γ_{d+1} such that $M_d = \partial\Gamma_{d+1}$. Under a gauge transformation $A_G \rightarrow A_G^\lambda$,

$$Z_{\text{bdry}}[A_G^\lambda] = Z_{\text{bdry}}[A_G]e^{i\alpha[A_G, \lambda]}, \quad Z_{\text{bulk}}[A_G^\lambda] = Z_{\text{bulk}}[A_G]e^{-i\alpha[A_G, \lambda]}. \quad (3)$$

We will not dwell on this general classification and those who are interested can refer to the original paper by Callan+Harvey [2], a modern physics-based discussion by Senthil [3], and a more mathematical treatment by Dan Freed [4].

Now that we have a complete definition of anomalies, we can state the main result that makes anomalies powerful.

Anomaly matching (informal): If a quantum system has a G -anomaly, then the ground state cannot be gapped, G -symmetric, and short-range-entangled. In other words, it is either (1) A gapped phase with long-range entanglement, (2) A phase that spontaneously breaks the G symmetry down to a smaller subgroup, (3) A gapless phase.

As promised, this result provides a non-perturbative constraint on the low energy (IR) physics from microscopic (UV) information (i.e. the 't Hooft anomaly of a global symmetry). Unfortunately, stating this result formally in complete generality is quite subtle and requires some amount of casework. In any case, since it is difficult to prove this result in general even with a careful definition, we will instead study specific examples and establish such a UV-IR connection in each setting. In all of our examples, the general flavor of the argument will be universal and will hopefully provide a solid foundation for further exploration

3 Minimal example in 0+1D: particle on a ring

The simplest anomalies that we will consider live in 0+1 dimensions. Remarkably, these examples will dispel several myths about anomalies that often appear in the literature:

1. Anomalies are associated with subtleties of quantum field theories and how the path integral is regularized in the UV.
2. Anomalies are associated with fermions.
3. A theory with a G anomaly in d dimensions is sick and ill-defined. It must be attached to a $d + 1$ dimensional bulk that “cancels” the anomaly.

While we have high energy physicists to blame for the first two myths, the last myth is entirely attributable to card-carrying condensed matter physicists. Finding published papers that contain the statements above is left as an exercise to the reader.

3.1 0+1D anomalies are projective representations

A 0+1D system is nothing but a system of finitely many quantum mechanical particles. A global symmetry G in quantum mechanics is represented by a unitary/anti-unitary operator U_g that commutes with the Hamiltonian for every $g \in G$ and acts faithfully on all the space of operators by conjugation $U(g)OU(g)^\dagger$. From this definition, we see that operators always transform in linear representations of G while states *can transform* in projective representations in G . This important observation has been made in Steve’s previous lectures.

Now recall the general definition of ’t Hooft anomalies: a system with symmetry G has a G -anomaly iff there is an obstruction to gauging the symmetry G . To determine whether or not such an obstruction is present, let us couple the system to a background G gauge field A (for simplicity, let us restrict to the case where G is a Lie group). In 0+1D, the coupling to A corresponds to the following modification of the Euclidean *phase space path integral*²

$$Z[A] = \int D\Phi e^{iS_B[\Phi] - \int d\tau H[\Phi] - i \int A_0^a(\tau) Q_a[\Phi] d\tau} . \quad (4)$$

Let us unpack this expression. Φ represents the phase space coordinates and A_0^a is the a -th component of the non-Abelian gauge field. $H[\Phi]$ is the classical Hamiltonian. The fields $Q_a[\Phi]$ are the non-Abelian conserved charges that generate the Lie algebra of G . $S_B[\Phi]$ is the Berry phase term that plays the role of $p\dot{q}$ for a general phase space and determines the commutation relations between operators upon quantization. The general procedure for quantizing a classical theory on an arbitrary phase space is known as geometric quantization. Interested readers can consult [5] for an elementary introduction geared towards physicists.

In the Hamiltonian representation, the same partition function can be written in a simpler way

$$Z[A] = \text{Tr} \left[e^{-i \int A_0^a(\tau) Q_a d\tau} e^{-\beta H} \right] , \quad (5)$$

where H is the quantum Hamiltonian and Q_a are the quantum conserved charge operators for G satisfying the commutation relations $[Q_a, Q_b] = i f_{abc} Q_c$ for some set of structure constants f_{abc} .

²The representation in configuration space path integral can always be derived from the phase space path integral.

In 0+1D, all configurations of A are gauge equivalent to flat configurations and the only gauge-invariant data are holonomies of A around the thermal circle S^1_β . In the Hamiltonian language, a flat gauge field implies $A_0^a(\tau)$ is independent of τ and the integral over τ can be trivially performed. We therefore make the replacement

$$e^{-i \int A_0^a(\tau) Q_a d\tau} \rightarrow e^{-i\beta A_0^a Q_a} . \quad (6)$$

For any constant choice of A_0^a , we can regard $e^{-i\beta A_0^a Q_a}$ as the (possibly projective) representation of a group element $h \in G$ in the quantum Hilbert space, which we denote as $U(h)$. The group element h is entirely determined by A_0^a , while the conserved charges Q_a are determined by the representation of the Lie algebra generators.

From this expression, we see that inserting a general holonomy labeled by $h \in G$ gives rise to a twisted partition function

$$Z[h] = \text{Tr} \left[U(h) e^{-\beta H} \right] . \quad (7)$$

Now suppose that the symmetry representation is projective with a projective phase $U(g_1)U(g_2) = \omega(g_1, g_2)U(g_1g_2)$. Then the partition function satisfies the identity

$$Z[h_1, h_2] = \text{Tr} \left[U(h_1)U(h_2)e^{-\beta H} \right] = \omega(h_1, h_2) \text{Tr} \left[U(h_1h_2)e^{-\beta H} \right] = \omega(h_1, h_2)Z[h_1h_2] . \quad (8)$$

The insertion of h_2 followed by h_1 is gauge-equivalent to the direct insertion of h_1h_2 . However, a nontrivial projective phase $\omega(h_1, h_2)$ implies that the partition function is **not invariant** under such a gauge transformation. This is precisely the definition of an 't Hooft anomaly!

3.2 Almost trivial example: a single spin- s

The minimal example that illustrates the general ideas described above is the quantum mechanics of a single spin- s degree of freedom. The phase space path integral for this system (see Problem 2) takes the form

$$Z_s = \int D\vec{n}(\tau) \exp \left\{ i s \cdot S_B[\vec{n}(\tau)] - \int_0^\beta d\tau H[s\vec{n}(\tau)] \right\} , \quad (9)$$

where β is the inverse temperature, H is the classical Hamiltonian and S_B is the Berry phase action

$$S_B[\vec{n}] = \int_0^\beta d\tau \int_0^1 du \vec{N}(\tau, u) \cdot \left[\partial_u \vec{N}(\tau, u) \times \partial_\tau \vec{N}(\tau, u) \right] , \quad (10)$$

where $\vec{N}(\tau, u)$ is an extension of $\vec{n}(\tau)$ onto $[0, \beta] \times [0, 1]$ such that $\vec{N}(\tau, 0) = \vec{n}_*$ for some choice of initial orientation $\vec{n}_*$ and $\vec{N}(\tau, 1) = \vec{n}(\tau)$. One can show that this action precisely measures the solid angle traced by the spin vector $\vec{n}(\tau)$ during the time interval $\tau \in [0, \beta)$ (see Problem 2). To maintain $SO(3)$ symmetry, we will choose $H = 0$. The entire spin coherent state path integral is therefore

$$Z_s = \int D\vec{n}(\tau) \exp \{ i s \cdot S_B[\vec{n}(\tau)] \} . \quad (11)$$

This is the world's most complicated way to compute the integer $2s + 1$, which is the dimension of the spin- s Hilbert space.

Now let us introduce an $SO(3)$ gauge field A . Following the general procedure and specializing to $s = 1/2$, the phase space path integral is modified to

$$Z_{1/2}[A] = \int D\vec{n}(\tau) \exp\left\{\frac{i}{2}S_B[\vec{n}] - i \int A_0^a(\tau) Q_a[\vec{n}] d\tau\right\}, \quad Q_a[\vec{n}] = \frac{1}{2}\vec{n}. \quad (12)$$

Let us now insert a 2π holonomy around S_β^1 for the $SO(3)$ gauge field A

$$A_\tau^1 = 2\pi/\beta, \quad A_\tau^2 = A_\tau^3 = 0. \quad (13)$$

As an $SO(3)$ gauge field, this configuration is related to the $A_\tau^a = 0$ configuration by a large gauge transformation $U(\tau) = e^{2\pi i \tau T_1/\beta}$ with T_1 being the first generator of the $SO(3)$ Lie algebra (see Problem 1). However, by explicit computation, we see that

$$Z_{1/2}[A] = \text{Tr } e^{-2\pi i Q_1} = -Z_{1/2}[0]. \quad (14)$$

The minus sign originates from the fact that $Q_1 = \frac{1}{2}\sigma_1$ has eigenvalue $\pm 1/2$ such that $e^{-2\pi i Q_1} = -1$. This -1 factor is a direct manifestation of the projective representation of $SO(3)$ on the spin-1/2 Hilbert space.

Therefore, the partition function is not invariant under a large $SO(3)$ gauge transformation, signaling an $SO(3)$ 't Hooft anomaly. You are encouraged to work through problem 2 and show that the anomaly disappears for $s \in \mathbb{Z}$, where the Hilbert space transforms in a linear rather than projective representation of $SO(3)$.

Before concluding this section, let us make a subtle remark. Consider a qutrit transforming in the spin-1 representation of $SO(3)$. Since this is linear representation, there is no 't Hooft anomaly and $Z_1[A]$ is perfectly gauge invariant. However, we also know that there is no $SO(3)$ invariant state in the qutrit Hilbert space. Since dynamical gauging is equivalent to the projection onto gauge-invariant states in 0+1D, we must conclude that gauging the $SO(3)$ produces a trivial theory. Indeed, integrating $Z_1[A]$ over all possible holonomies of A around S_β^1 produces a partition function $Z_{1,\text{gauged}} = 0$. This example illustrates that there is a difference between “anomaly of a global symmetry G ” and “triviality of the G -gauged theory”. The first condition says that the partition function $Z[A]$ in the presence of a background gauge field A fails to be gauge invariant, while the second condition says that integrating $Z[A]$ over all A produces a trivial result.

3.3 Minimal nontrivial example: particle on a ring

The spin- s example is nice and clean conceptually. However, it feels a bit useless. We have not learned much physics by figuring out the 't Hooft anomaly. To really convince you of the power of anomalies, let us consider a slightly more complicated example: a single quantum mechanical particle on a ring of radius 1 described by a one-parameter (θ) family of Hamiltonians [6]

$$H(\theta) = \frac{1}{2}\left(p - \frac{\theta}{2\pi}\right)^2, \quad [q, p] = i, \quad q \sim q + 2\pi. \quad (15)$$

From this Hamiltonian, we can easily pass to a path integral description in Euclidean time, governed by the Euclidean action³

$$S_E[q] = \frac{1}{2} \int d\tau \dot{q}^2 + \frac{i}{2\pi} \int d\tau \theta \dot{q}. \quad (16)$$

When $\theta = 0/\pi$, there is a $\mathbb{Z}_2^C$ charge conjugation symmetry $q \rightarrow -q$ in addition to the $U(1)$ symmetry $q \rightarrow q + \alpha$. If we denote the charge conjugation element by c and a $U(1)$ shift of α by $u(\alpha)$. Then it is clear that $cu(\alpha)c = u(-\alpha)$. Thus the full symmetry group is a semi-direct product $O(2) = U(1) \rtimes \mathbb{Z}_2^C$.

Now let us ask what happens when we turn on a background $U(1)$ gauge field A_0 , which amounts to replacing $\dot{q}$ with $\dot{q} - A_0$. The Euclidean action transforms to

$$S_E[q, A] = \frac{1}{2} \int d\tau (\dot{q} - A_0)^2 + \frac{i}{2\pi} \int d\tau \theta (\dot{q} - A_0). \quad (17)$$

When $\theta = 0$, we retain the $\mathbb{Z}_2^C$ symmetry under which $q \rightarrow -q, A \rightarrow -A$. However, it is clear that for $\theta = \pi$, the action $S_E[A]$ is not invariant under the same transformation. To see the mismatch, let us compute the partition functions

$$Z_\theta[A] = \int Dq e^{-S_E[q, A]} = \int Dq e^{-S_E[q, -A]} \cdot e^{i\frac{\theta}{\pi} \int A_0 d\tau} = Z_\theta[-A] \cdot e^{i\frac{\theta}{\pi} \int A_0 d\tau}. \quad (18)$$

For $\theta = \pi$, the two partition functions differ by precisely the elementary Wilson loop of A

$$Z_\pi[A] = Z_\pi[-A] \cdot e^{i \int A_0 d\tau}. \quad (19)$$

Furthermore, it is easy to see that this difference cannot be removed by any local counterterms. In 0+1D, the only available local counterterm is $e^{ik \int A_0 d\tau}$ for $k \in \mathbb{Z}$. If we attach such a counterterm and define $\tilde{Z}_\pi[A] = Z_\pi[A] e^{ik \int A_0 d\tau}$, then the new partition function satisfies

$$\tilde{Z}_\pi[A] = \tilde{Z}_\pi[-A] \cdot e^{i(2k+1) \int A_0 d\tau}. \quad (20)$$

For any integer k , the phase cannot be canceled, and we conclude that there is a genuine 't Hooft anomaly.

By the general correspondence between 't Hooft anomalies and projective representations in 0+1D, we should also be able to deduce the anomaly directly from the projective representation of $O(2) = U(1) \rtimes \mathbb{Z}_2^C$. To see the projective representation, let us solve the spectrum explicitly at $\theta = \pi$. The Hamiltonian and its eigenstates are

$$H(\pi) = \frac{(p - 1/2)^2}{2}, \quad H|n\rangle = E_n|n\rangle, \quad E_n = \frac{(n - 1/2)^2}{2}, \quad \langle q|n\rangle = \frac{1}{\sqrt{2\pi}} e^{inq}. \quad (21)$$

³Here we will use the configuration space path integral just to illustrate that it produces the same answer as the general procedure in the previous section based on phase space path integrals.

At the level of operators, charge conjugation C acts as $q \rightarrow -q$ and $p \rightarrow -p + 1$ while $U(\alpha)$ shifts $q \rightarrow q + \alpha$ while leaving p invariant. Acting on the states, one can check that C sends $|n\rangle$ to $|-n + 1\rangle$ (since n is basically the momentum) while $U(\alpha)|n\rangle = e^{i\alpha n}$. If we now compute the action of $CU(\alpha)C$ on an eigenstate, we see that

$$CU(\alpha)C|n\rangle = e^{i\alpha(-n+1)}C|-n+1\rangle = e^{-i\alpha n}e^{i\alpha}|n\rangle = e^{i\alpha}U(-\alpha)|n\rangle. \quad (22)$$

Due to the factor $e^{i\alpha}$, we conclude that the representation of the $O(2)$ group on the Hilbert space is indeed projective and the $O(2)$ group has an 't Hooft anomaly.

3.4 Anomaly-protected degeneracy

Given the discussion so far, one may reasonably complain: you have given me a free Hamiltonian which can be exactly solved. What is the point of introducing anomalies?

To answer this question, let us now perturb the Hamiltonian with an interaction term $V \cos(2q)$ which breaks the $U(1)$ symmetry down to a $\mathbb{Z}_2$ subgroup $q \rightarrow q + \pi$

$$H(\theta, V) = \frac{(p - 1/2)^2}{2} + V \cos(2q). \quad (23)$$

The spectrum of this interacting quantum mechanics problem *cannot be solved exactly* anymore. Nevertheless, we know from our previous analysis that the charge conjugation operator must act by sending $p \rightarrow -p + 1$ and $q \rightarrow -q$. The same calculation as before shows that $CU = -UC$ when acting on all states. Clearly, all states must then transform in a projective representation of $\mathbb{Z}_2 \times \mathbb{Z}_2^C$, while operators transform in a linear representation of $\mathbb{Z}_2 \times \mathbb{Z}_2^C$. This $\mathbb{Z}_2 \times \mathbb{Z}_2^C$ anomaly prevents the deformation V from removing the 2-fold ground state degeneracy. This argument is the simplest instantiation of 't Hooft anomaly matching and illustrates their power in producing **non-perturbative constraints** on the low energy spectrum of an interacting quantum system.

4 Example in 1+1D: Luttinger liquid and the axial anomaly

Now let us upgrade to 1+1 dimensions, where anomalies provide more striking constraints. A particularly interesting class of theories are compact boson conformal field theories (possibly coupled to dynamical $\mathbb{Z}_2$ gauge fields). To high energy physicists, these theories are interesting because compact bosons are target space coordinates in worldsheet string theory. To condensed matter physicists, these theories arise naturally as low energy descriptions of 1+1D interacting electronic systems, sometimes referred to as Luttinger liquids. From the electronic perspective, these Luttinger liquids go beyond the celebrated Fermi liquid paradigm, as they do not contain well-defined electronic quasiparticles. We will see how much we can understand these systems from the perspective of 't Hooft anomalies.

4.1 Deriving the axial anomaly

To set up the problem, let us begin with a partially filled band in 1+1 dimensions. Given an arbitrary electron dispersion $\epsilon(k)$ and a chemical potential μ , the low energy description of the partially filled band can be obtained by focusing on fermion creation and annihilation operators near the momenta k_R/k_L that separates occupied from unoccupied states on the right/left edge of the filled momentum eigenstates. This expansion amounts to the operator decomposition

$$c(x) = e^{ik_R x} \psi_R(x) + e^{ik_L x} \psi_L(x). \quad (24)$$

In order for this decomposition to be well-defined, we must restrict to smooth modes of ψ_R, ψ_L with wavelength $\lambda \gg 1/|k_R - k_L|$. This restriction makes sense only in the low energy limit. From now on, we will focus on the case in which the Fermi velocity at the two endpoints k_R, k_L are equal and opposite (this can be guaranteed by imposing e.g. time reversal symmetry). With an appropriate choice of units, the Fermi velocity can be set to ± 1 . With this choice, these chiral fields combine into a Dirac spinor ψ , whose quadratic action takes the form

$$S_0 = \int \bar{\psi} [\gamma^0 \partial_\tau + v_F \gamma^x \partial_x] \psi, \quad \gamma^0 = \sigma^x, \quad \gamma^1 = -\sigma^y, \quad \gamma^5 = i\gamma^0 \gamma^1 = \sigma^z. \quad (25)$$

When $k_R - k_L$ is incommensurate with the lattice, translation symmetry forbids scattering events that take electrons from the left to the right edge and vice versa. Therefore, the low energy theory enjoys an emergent $U(1)_R \times U(1)_L$ symmetry that rotates the chiral fields ψ_R, ψ_L separately. Taking linear combinations of the chiral $U(1)$ symmetries gives

$$U(1)_v : \psi_{R/L} \rightarrow e^{i\alpha} \psi_{R/L}, \quad U(1)_a : \psi_{R/L} \rightarrow e^{\pm i\beta} \psi_{R/L}. \quad (26)$$

The first symmetry $U(1)_v$ is the familiar $U(1)$ symmetry associated with microscopic electron number conservation. The second symmetry $U(1)_a$ is an axial rotation symmetry which is *emergent* in the low energy theory. The conserved currents associated with these symmetries are

$$J_v^\mu = \bar{\psi} \gamma^\mu \psi, \quad J_a^\mu = \bar{\psi} \gamma^\mu \gamma^5 \psi. \quad (27)$$

Following the standard procedure for detecting 't Hooft anomalies, let us now attempt to couple the theory to background gauge fields A_v and A_a and see if the resulting action is gauge invariant. In the Euclidean action for Dirac fermions, we can simply perform minimal coupling and replace ∂_μ with a covariant derivative $\partial_\mu - iA_{v,\mu} - i\gamma^5 A_{a,\mu}$. The resulting action is

$$S_\psi[A_a, A_v] = \int \bar{\psi} \gamma^\mu [\partial_\mu - iA_{v,\mu} - i\gamma^5 A_{a,\mu}] \psi. \quad (28)$$

To derive the effective action in the presence of the background gauge field, we can integrate out the fermion fields

$$Z[A_a, A_v] = \int D\bar{\psi} D\psi \exp\{-S_\psi[A_a, A_v]\} = e^{-S_{\text{eff}}[A_a, A_v]}. \quad (29)$$

This integration involves a Feynman diagram calculation (in the path integral formalism) or a point-splitting analysis (in the canonical quantization formalism), which I will leave to a standard quantum field theory class (see [7] for a remarkably succinct one-page derivation). We will actually take a different approach that circumvents this loop integration entirely.

The basic idea behind this alternative approach is that in 1+1D, the Dirac fermion theory admits an equivalent representation in terms of a compact boson coupled to a **dynamical $\mathbb{Z}_2$ gauge field**. The mapping from the Dirac fermion theory to the compact boson theory is often referred to as “bosonization”. Although the dynamical $\mathbb{Z}_2$ gauge field is indispensable when the theory is quantized on arbitrary closed manifolds (see [8] for a modern treatment), it can be set to zero when the theory is quantized on an infinite line. In this lecture, we will restrict our attention to the infinite line, which is sufficient for most practical applications.

The operator mapping in bosonization takes the schematic form

$$\psi_R^\dagger \rightarrow \frac{1}{\sqrt{2\pi x_c}} e^{i(\phi+\theta/2)}, \quad \psi_L^\dagger \rightarrow \frac{1}{\sqrt{2\pi x_c}} e^{i(\phi-\theta/2)}, \quad \rho_R + \rho_L \rightarrow \frac{1}{2\pi} \partial_x \theta. \quad (30)$$

The $U(1)_v$ and $U(1)_a$ symmetries act on the compact scalars ϕ, θ as

$$U(1)_v : \phi \rightarrow \phi + \alpha, \quad U(1)_a : \theta \rightarrow \theta + 2\beta. \quad (31)$$

Since ϕ, θ are both 2π -periodic, we see that the faithful symmetry is not $U(1)_v \times U(1)_a$ but rather $U(1)_v \times U(1)_a / \mathbb{Z}_2$.

The bosonization dictionary maps the free Dirac theory to a compact boson theory with partition function

$$Z_{\theta,\phi} = \int D\theta D\phi \exp\{-S_{\theta,\phi}\}, \quad S_{\theta,\phi} = \int \frac{i}{2\pi} \partial_x \theta \partial_t \phi + \frac{1}{2\pi} \left[\frac{1}{4} (\partial_x \theta)^2 + (\partial_x \phi)^2 \right]. \quad (32)$$

Remarkably, note that the leading order four-fermion interaction $g(\rho_R + \rho_L)^2$ maps to a quadratic deformation of the compact boson theory $\frac{g}{4\pi^2} (\partial_x \theta)^2$. Therefore, an interacting Dirac action with coupling g maps to a compact boson action

$$S_{\theta,\phi}[g] = \int \frac{i}{2\pi} \partial_x \theta \partial_t \phi + \frac{1}{2\pi} \left[\left(\frac{1}{4} + \frac{g}{2\pi} \right) (\partial_x \theta)^2 + (\partial_x \phi)^2 \right]. \quad (33)$$

Upon integrating out the θ field in favor of the ϕ field and appropriately rescaling the spatial coordinate (see Problem 3), we land on the canonical representation of the compact boson partition function

$$Z_\phi[g] = \int D\phi \exp\{-S_\phi[g]\}, \quad S_\phi[g] = \frac{R(g)^2}{4\pi} \int dt dx (\partial_\mu \phi)^2, \quad R(g)^2 = \frac{2}{1 + \frac{2g}{\pi}}. \quad (34)$$

While a proof of this duality is beyond the scope of this lecture, Problem 3 walks you through some checks of this duality that should convince you that it is not totally crazy. You can refer to the classic textbook [9] for a more complete treatment.

Armed with this duality, we can derive the $U(1)_v \times U(1)_a$ anomaly within the compact boson theory and then directly transfer our results to the original fermionic problem. To derive the anomaly in the compact boson theory, we first identify the $U(1)_v$ and $U(1)_a$ currents in Euclidean signature

$$J_v^\mu = -i \frac{R^2}{2\pi} \partial_\mu \phi, \quad J_a^\mu = \frac{1}{2\pi} \epsilon_{\mu\nu} \partial^\nu \phi. \quad (35)$$

Once we identify the currents, we can couple the compact boson field theory to background A_v and A_a gauge fields

$$S_\phi[A_v, A_a] = \int d^2x \left[\frac{R^2}{4\pi} (\partial_\mu \phi - A_{v,\mu})^2 + \frac{i}{2\pi} \epsilon_{\mu\nu} (\partial_\nu \phi - A_{v,\nu}) A_{a,\mu} \right]. \quad (36)$$

Under a simultaneous $U(1)_v$ and $U(1)_a$ gauge transformation parametrized by λ_v, λ_a , the fields transform as

$$\phi \rightarrow \phi + \lambda_v, \quad A_{v,\mu} \rightarrow A_{v,\mu} + \partial_\mu \lambda_v, \quad A_{a,\mu} \rightarrow A_{a,\mu} + \partial_\mu \lambda_a. \quad (37)$$

Plugging this transforming into the action, we see that

$$S_\phi[A_v, A_a] \rightarrow S_\phi[A_v, A_a] - \frac{i}{2\pi} \int d^2x \lambda_a (\partial_\tau A_{v,x} - \partial_x A_{v,0}). \quad (38)$$

This additional phase factor proportional to λ_a cannot be absorbed by a local counterterm that depends only on the background fields A_v and A_a (see Problem 3). From this transformation law, we immediately deduce an anomalous current conservation law for the $U(1)_a$ symmetry

$$\partial_\mu J_a^\mu = \frac{1}{2\pi} E, \quad E = -\partial_\tau A_{v,x} + \partial_x A_{v,0}. \quad (39)$$

From the perspective of the original Dirac fermion theory, this 't Hooft anomaly implies that upon **dynamically gauging** the $U(1)_v$ symmetry, the $U(1)_a$ symmetry must be explicitly broken. This is nothing but the celebrated ABJ anomaly for 1+1D QED, in which the $U(1)_v$ gauge field is a dynamical gauge field to begin with!

Remarkably, the $U(1)_v \times U(1)_a$ 't Hooft anomaly does not depend on the compact boson radius R . If we recall the relation $R(g)^2 = \frac{2}{1+2g/\pi}$, we immediately conclude that the anomaly is *independent of* the strength of four-fermion interactions. Even without the ability to solve the interacting fermion problem exactly, we can use its 't Hooft anomaly to derive non-perturbative constraints. In the rest of this section, we will see two examples of such constraints in 1+1D Luttinger liquids. Both of these constraints can be generalized to a large class of 2+1D non-Fermi liquids, as shown in [10, 11, 12].

4.2 Consequence of the anomaly I: Exact results on transport

An immediate implication of the anomaly is that electrical transport can be computed exactly. Let us see how this works. From the definition of $U(1)_v$ and $U(1)_a$ currents, we know that

$$J_v^\tau = iR^2 J_a^x, \quad J_v^x = -iR^2 J_a^\tau. \quad (40)$$

In other words, the charge current is the chiral density, and the chiral current is the charge density. From the anomaly equation, we also know that

$$\partial_\mu J_a^\mu = \frac{1}{2\pi}(\partial_x A_{v,t} - \partial_t A_{v,x}) = \frac{1}{2\pi}E, \quad (41)$$

where E is the electric field in 1+1D. Specializing to the spatially uniform conductivity, we perform a Fourier transform from (τ, x) to (ω, q) and set $q = 0$. The resulting equation simplifies to

$$-i\omega J_a^\tau(\omega, q = 0) = \frac{1}{2\pi}E(\omega, q = 0). \quad (42)$$

Finally, using the relationship between J_a^t and J_v^x , we find an exact formula for the conductivity

$$J_v^x(\omega, q = 0) = -iR^2 J_a^t(\omega, q = 0) = \frac{R^2}{2\pi\omega}E(\omega, q = 0). \quad (43)$$

Upon analytically continuing back to real frequencies $i\omega \rightarrow \Omega + i0^+$, we can extract the physical conductivity

$$\sigma(\Omega + i0^+, q = 0) = \frac{iR^2}{2\pi(\Omega + i0^+)}. \quad (44)$$

We emphasize that this is a non-perturbative result with only a single unknown parameter R that depends on the detailed form of UV interactions. The functional form of $\sigma(\Omega)$ as a function of Ω is *completely fixed* by the 't Hooft anomaly. Since R can be directly measured from the electron Green's function, we can also interpret the anomaly as an exact relation between the electron Green's function and the uniform conductivity in a 1+1D electronic metal.

4.3 Consequence of the anomaly II: Luttinger's theorem

To further illustrate the power of this 't Hooft anomaly, let us use it to prove a foundational result in condensed matter theory known as Luttinger's theorem, following the argument in [10]. The basic statement of Luttinger's theorem, valid in any spatial dimension, is that the volume of the Fermi sea V_F in a metal (made of spinless fermions) is related to the filling fraction ν (i.e. charge per unit cell) by a universal relation

$$\nu = \frac{V_F}{(2\pi)^d} \mod 1. \quad (45)$$

This theorem is a beautiful example of a UV-IR relation: the charge filling ν is defined by the Hamiltonian while the Fermi sea volume V_F is a notion that only makes sense in the IR limit within a metallic phase. From this perspective, Luttinger's theorem provides a non-perturbative constraint on the IR theory directly in terms of UV data.

To prove this theorem in 1+1D, let us recall the embedding of the UV lattice translation symmetry in the IR. Focusing on the two Fermi points k_R and k_L , it is obvious that the total

momentum carried by the electrons is given by $k_R N_R + k_L N_L$ where $N_{R/L}$ are the number of right and left moving electrons. As a result, the lattice translation operator $T = e^{iP}$ embeds in the IR as

$$\rho(T) = e^{i(k_R N_R + k_L N_L)}. \quad (46)$$

Now let us couple the system to a background $U(1)$ gauge field A and thread a 2π holonomy of A_x around the spatial circle $\int dx A_x = 2\pi$. We will compute the response of the system to this background gauge field, first in the microscopic lattice model (UV) and then in the low energy interacting Dirac fermion field theory (IR).

In the UV, the holonomy of A_x is implemented by a large gauge transformation

$$U_{2\pi} = e^{i \frac{2\pi i}{L} \sum_{x=0}^{L-1} x Q_x}, \quad (47)$$

where Q_x is the $U(1)$ charge operator at site x . Now recall that $T Q_x T^{-1} = Q_{x+1}$. Using this commutation relation, it is easy to show that

$$U_{2\pi} T U_{2\pi}^{-1} = T e^{2\pi i \nu}, \quad (48)$$

where ν is the charge filling fraction, defined mod 1.

Now let us compute the same response using the IR Dirac fermion field theory. The $U(1)$ symmetry in the UV embeds as the $U(1)_v$ symmetry. Due to the 't Hooft anomaly, we know that 2π flux of $U(1)_v$ traps one unit of axial charge, increasing N_R by 1 and decreasing N_L by 1. As a result, the IR translation eigenvalue shifts to

$$e^{iP} \rightarrow e^{iP} \cdot e^{i(k_R - k_L)}. \quad (49)$$

Comparing the results of these two calculations in the UV and in the IR, we conclude that

$$\nu = \frac{k_R - k_L}{2\pi} \mod 1. \quad (50)$$

Since $k_R - k_L$ is precisely the definition of Fermi sea volume in one spatial dimension, we have proven the 1+1D Luttinger's theorem. This proof is conceptually clean and remarkably simple, especially if you have previously encountered the old-fashioned proof using Luttinger-Ward functionals constructed out of many-body Green's functions.

5 Modern developments

The discussion of anomalies in 0+1D and 1+1D has hopefully convinced you that 't Hooft anomalies encode interesting non-perturbative structures of many-body quantum systems and provide powerful constraints on low energy dynamics. There are many modern research directions related to 't Hooft anomalies that we did not get a chance to cover. The purpose of this last section is to give you a flavor of these directions and provide some relevant references.

5.1 Anomalies of spacetime symmetries

In this lecture, we have exclusively focused on anomalies of internal symmetries, i.e. symmetries that do not act on the underlying spacetime. This is precisely the setting in which 't Hooft originally defined anomalies. However, many lattice models and quantum field theories have spacetime symmetries such as translation/rotation. It is not obvious what it means to “dynamically gauge” such spacetime symmetries, and the definition of 't Hooft anomalies in this lecture does not generalize in a straightforward way.

Nevertheless, since spacetime symmetries are ubiquitous in the UV, we better find some way to use them to constrain the IR dynamics. In fact, many famous constraints exist in the literature:

Constraint from translation alone [15]: Any eigenstate of lattice translation e^{iP} with eigenvalue $e^{iP} \neq 1$ must be long-range entangled (i.e. $|\Psi\rangle$ cannot be transformed to an unentangled product state through an adiabatic evolution or a finite-depth quantum circuit). As a corollary, a 1d translation and $U(1)$ symmetric state at irrational $U(1)$ charge filling must be long-range entangled.

Lieb-Schultz-Mattis-Oshikawa-Hastings [16, 17, 18]: A spin 1/2 chain with $SO(3)$ and lattice translation symmetry cannot have a unique gapped ground state. More generally, a translation-invariant, G -symmetric local Hamiltonian with a nontrivial projective representation of G per unit cell cannot have a non-degenerate, gapped ground state.

Constraint on charge fluctuation [20]: Consider a spin model with $SO(3)$ symmetry and lattice translations. Define $F(a, b) = \left\langle \left(\sum_{j \in [a, b]} S_j^z \right)^2 \right\rangle$ as the variance of the total S^z in an interval $[a, b]$. In the thermodynamic limit, if $F(a, b)$ is bounded as $|b - a|$ increases, the ground state must spontaneously break the lattice translation symmetry.

These theorems smell very similar to 't Hooft anomaly matching: some UV information constrains the IR phase to be nontrivial. A growing number of papers are attempting to interpret these theorems as consequences of a mixed 't Hooft anomaly between spatial and internal symmetries. A concise summary of this general program can be found in a recent review article [19].

5.2 The relationship between lattice anomalies and QFT anomalies

Many interesting models in condensed matter physics are lattice models defined on a finite d -dimensional spatial lattice. The low energy limit of these lattice models can often be approximated by continuum quantum field theories in the same spatial dimension. Therefore, it is natural to expect that there is a connection between 't Hooft anomalies on the lattice and 't Hooft anomalies in the continuum QFT. At the very least, one should expect that there is a map ρ from anomalies

on the lattice to anomalies of the QFT.

It is natural to ask: is the map ρ surjective, injective, both, or neither? This question turns out to be incredibly subtle, as highlighted by two recent papers [13, 14]. Assuming that the onsite Hilbert space dimension is finite, then under the definition of lattice anomalies put forth in [14], the map ρ is neither surjective nor injective. This means that there are lattice anomalies that map to trivial QFT anomalies, and there are nontrivial QFT anomalies with no lattice origin. It remains to be seen whether allowing for infinite-dimensional onsite Hilbert space dimension will change these conclusions.

Even in cases where a nontrivial lattice anomaly maps into a nontrivial QFT anomaly, the mapping could be quite subtle. Sometimes, a lattice symmetry in the UV (e.g. translation/reflection) can map into an internal symmetry in the IR. In these cases, a mixed anomaly in the UV between spatial and internal symmetries can be matched by a mixed anomaly between internal symmetries in the IR [21]. In more exotic cases, anomalies of ordinary global symmetries in the UV can be matched by anomalies of higher-form/generalized symmetries in the IR. A complete framework for the most general lattice-QFT anomaly matching remains to be developed.

5.3 Symmetry-enforced gaplessness

In all of our examples so far, a nontrivial 't Hooft anomaly forbids a trivial phase but leaves many options open. For example, in two spatial dimensions, there are generally three types of anomaly-allowed IR phases: (1) gapped topological orders, (2) spontaneous symmetry breaking phases, and (3) gapless phases. It is natural to ask: are there situations in which the constraint can be further strengthened? In other words, can we use symmetries and anomalies in a more clever way to rule out one of the three possibilities?

This question was posed and answered affirmatively by [22] in a specific context: the surface state of a 3+1D topological superconductor in class CI with index $k = 1$. Here we briefly review their argument. By definition, the symmetry that protects a class CI topological superconductor is $SU(2) \times \mathbb{Z}_2^T$, where $SU(2)$ is a fermionic spin conservation symmetry and $\mathbb{Z}_2^T$ is an anti-unitary time reversal symmetry. For a free-fermion state with index $k \in \mathbb{Z}$, there are $2k$ Dirac cones on the surface, realizing k flavors of $SU(2)$ fundamental fermions. The corresponding free fermion Hamiltonian is

$$\sum_{i=1}^k \psi_i^\dagger (p_x \sigma_x + p_y \sigma_z) \otimes \tau_0 \psi_i, \quad (51)$$

where τ_μ are Pauli matrices that carry $SU(2)$ spin indices and σ_μ are Pauli matrices that carry Dirac spin indices. The global $SU(2)$ symmetry acts as $\psi_i \rightarrow \sigma_0 \otimes U \psi_i$, while time-reversal acts as $\psi_i \rightarrow i \sigma_y \otimes \tau_0 \psi_i^\dagger$.

When $k = 1$, it is easy to check that one cannot write a quadratic gap term in the Hamiltonian that breaks $U(1)$ but not $SU(2)$. In fact, something much stronger is true: even with interactions, it is impossible for the surface theory to realize a symmetric gapped topological order at all. To see that, suppose that a symmetric gapped topological order is indeed realized at the $k = 1$ surface.

Since $SU(2)$ has no projective representations, every anyon in the topological order must transform in a linear irreducible representation of $SU(2)$. Furthermore, since the electron c transforms in the spin-1/2 representation, we can always bind an integer number of electrons to a quasiparticle to make an $SU(2)$ singlet. The topological order thus factorizes as $\{1, \epsilon, \dots\} \times \{1, c\}$ where $\{1, \epsilon, \dots\}$ is a purely bosonic topological order with $\mathbb{Z}_2^T$ symmetry and $SU(2)$ -singlet anyons. The 3+1D bulk theory of this 2+1D bosonic surface state must therefore be a bosonic SPT protected by $\mathbb{Z}_2^T$ symmetry (this step is based on anomaly inflow, which we briefly mentioned but did not treat in any detail). Moreover, by stacking two copies of the $k = 1$ topological superconductor together, we obtain the $k = 2$ topological superconductor which is also a bosonic SPT. Since bosonic SPTs in 3+1D with $\mathbb{Z}_2^T$ symmetry are classified by $\mathbb{Z}_2^3$, no SPT admits a “square root”. This means that the assumption of the $k = 1$ surface state being a symmetric gapped topological order must be wrong.

In summary, using the structure of symmetries and anomalies in the surface state in class CI topological superconductors, we reach the strong conclusion that the only possible IR phases are either gapless or spontaneously break the $SU(2) \times \mathbb{Z}_2^T$ symmetry. If the symmetry is preserved, then gaplessness becomes inevitable. This phenomenon was dubbed “symmetry-enforced gaplessness” [22].

This phenomenon has found generalizations in many other contexts, including discrete symmetries and even non-anomalous continuous symmetries [23]. However, the most general formal theoretical framework is not as well-developed as ’t Hooft anomalies in general. It would be extremely interesting to find more examples and make progress towards formulating a general principle.

5.4 Anomalies of infinite-dimensional symmetries

Finally, let us make lives harder by introducing infinite-dimensional symmetries. While symmetries of microscopic models of condensed matter systems are always described by finite-dimensional groups, the low energy effective theories of these systems could enjoy an infinite dimensional emergent symmetry (i.e. infinitely many conserved charges in the thermodynamic limit). The simplest example is the infinite-dimensional Virasoro symmetry of 1+1D conformal field theories. While the global conformal group is finite-dimensional, the local conformal algebra is an infinite-dimensional Virasoro algebra. This infinite-dimensional algebra is an important part of the reason why many 1+1D conformal field theories can be exactly solved.

Going beyond 1+1D, another interesting infinite-dimensional symmetry arises in 2+1D metallic systems, the simplest example being the 2+1D Landau Fermi liquid. In the uniform limit, the Landau Fermi liquid is described by a Hamiltonian

$$H = \int d\theta \epsilon(\theta) n_\theta + \int d\theta d\theta' n_\theta n_{\theta'} F(\theta, \theta'), \quad (52)$$

where θ, θ' label angles along the Fermi surface, n_θ is the charge density on the Fermi surface, and $F(\theta, \theta')$ are Landau parameters. Clearly, H commutes with n_θ for all θ and the angular charge density n_θ is a conserved quantity in the IR theory that counts the density of Landau quasiparticle excitations at angle θ . The group generated by these charge densities is known as the loop group

$LU(1)$ with group elements

$$LU(1) = \{e^{i \int d\theta f(\theta) n_\theta} | f : U(1) \rightarrow U(1)\}. \quad (53)$$

Note that the constant functions $f(\theta) = \alpha$ correspond to the $U(1)$ subgroup of $LU(1)$, which describes the microscopic charge conservation symmetry.

Remarkably, when we go beyond Landau Fermi liquids and study non-Fermi liquids, the Landau quasiparticles may be destroyed but the infinite-dimensional $LU(1)$ symmetry survives! Of course, the conserved charges n_θ no longer counts quasiparticles. Following the general philosophy of anomaly matching, we expect that 't Hooft anomalies of the $LU(1)$ symmetry should place strong constraints on the IR dynamics of non-Fermi liquids. This general research program was initiated in [10], where it was shown that many foundational results in the theory of metals such as Luttinger's theorem and quantum oscillations are direct consequences of the $LU(1)$ anomaly. Further developments in the past few years utilized the $LU(1)$ anomaly to derive exact thermodynamic susceptibilities, transport observables, as well as neutral collective modes in non-Fermi liquids. This research program is far from complete and there are many challenges and puzzles as reviewed in [24]. A deeper understanding of symmetries and anomalies in non-Fermi liquids will likely yield new insights about the structure of strongly interacting metals, which remains one of the deepest unsolved problems in condensed matter physics.

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6 Exercises

Problem 1: For a general non-Abelian gauge field A_0^a in 0+1D, a general gauge transformation generated by a time-dependent unitary matrix $U(\tau)$ acts on the gauge field as

$$A_0 \rightarrow U(\tau)^{-1} A_0 U(\tau) - iU(\tau)^{-1} \partial_\tau U(\tau). \quad (54)$$

Show that $U(\tau) = e^{2\pi i \tau T^1 / \beta}$ induces a gauge transformation

$$A_0^1 \rightarrow A_0^1 + 2\pi/\beta, \quad A_0^{a \neq 1} \rightarrow A_0^{a \neq 1}. \quad (55)$$

Problem 2: Consider a single spin- s quantum mechanical degree of freedom governed by the classical Hamiltonian H .

1. Derive the Euclidean spin coherent state path integral for this system on a thermal circle S_β^1 . Your final answer should agree with Eq. (9).
2. Prove that the Berry phase $e^{isS_B[\vec{n}]}$ is independent of the extension $\vec{n}(\tau) \rightarrow \vec{N}(\tau, u)$ whenever $2s \in \mathbb{Z}$.
3. In the lecture, we focused on the case of spin $1/2$. Generalize the argument in the lecture to show that there is no $SO(3)$ 't Hooft anomaly for $s \in \mathbb{Z}$, but there is an anomaly for $s \in \mathbb{Z} + 1/2$.

Problem 3: In this problem, we fill in some details in the discussion of 1+1D compact boson CFTs.

1. Start with the action $S_{\theta, \phi}[g]$, integrate out the θ field to derive the action $S_\phi[g]$, which depends on a single parameter $R(g)$. What happens if you instead integrate out ϕ to get an effective action $S_\theta[g]$? Show that

$$S_\theta[g] = \int \frac{\tilde{R}^2}{4\pi} (\partial_\mu \theta)^2, \quad \tilde{R} = 1/R. \quad (56)$$

2. In general, the partition function $Z[A_v, A_a]$ can be modified by local counterterms that depend on the background fields A_v, A_a . Write down the most general such counterterm and show that they do not remove the 't Hooft anomaly.
3. Compute the two-point correlation function of the vertex operator $e^{i\phi \pm i\theta/2}$ in the theory defined by $Z_{\theta, \phi}[g]$. Show that for $R = \sqrt{2}$, they reproduce the two-point correlation functions of ψ_R, ψ_L in the free Dirac fermion theory.